

Myeongseok Ryu

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Myeongseok Ryu is under Ph.D. program. (compiled on September 15, 2026)

RESEARCH INTERESTS

Control Theory

Adaptive Control, Optimal Control

Neural Network-based Control

Neuro-Adaptive Control, Reinforcement Learning

Contraction Theory

Vehicle Dynamics and Control

Active Rear Steering (ARS) Control

PROFILE & LOGOS



KAIST

MIC Lab
Mobility Intelligence and Control Laboratory

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST), Korea

September 2025 – Present

CCS Graduate School of Mobility

Ph.D. of Science in Mobility Engineering

- Supervisor: Prof. Kyunghwan Choi, KAIST

Gwangju Institute of Science and Technology (GIST), Korea, (Withdrew for further studies)

March 2025 – May 2025

School of Mechanical Engineering

Ph.D. of Science in Mechanical Engineering

Gwangju Institute of Science and Technology (GIST), Korea

March 2023 – February 2025

School of Mechanical Engineering

Master of Science in Mechanical Engineering

- Thesis: Constrained Optimization-Based Neuro-Adaptive Control (CoNAC) for Euler-Lagrange Systems
- Supervisor: Prof. Kyunghwan Choi, GIST

Incheon National University (INU), Korea

March 2017 – February 2023

Department of Mechanical Engineering

Bachelor of Engineering

PROFESSIONAL EXPERIENCE

Korea Advanced Institute of Science and Technology (KAIST), Korea

May 2025 – August 2025

Researcher, MIC Lab, CCS Graduate School of Mobility

– Research on Neural Network-based Control for Mobility Systems

RESEARCH PROJECTS

Development of a Rear-Wheel Steering (RWS) Control Algorithm for Improved Vehicle Handling Performance and Tunability (Phase II)

September 2026 – August 2027

Sponsor: Hyundai Motor Company (HMC)

Role: Lead Researcher

– Research on Active Rear Steering (ARS) Control for Vehicle Lateral Dynamics

Development of a Rear-Wheel Steering (RWS) Control Algorithm for Improved Vehicle Handling Performance and Tunability (Phase I)

August 2025 – July 2026

Sponsor: Hyundai Motor Company (HMC)

Role: Lead Researcher

– Research on Active Rear Steering (ARS) Control for Vehicle Lateral Dynamics

SKILLS

Languages: Korean, English
Programming: Matlab/Simulink, Python, C/C++
Implementation: **Simulation** CarMaker, ROS
Others Git, LaTeX, Jekyll

PUBLICATIONS

Papers Under Review

2. Constrained Optimization-Based Neuro-Adaptive Control (CONAC) for Unknown Systems Under Multiple Convex Input Constraints
Myeongseok Ryu, Donghwa Hong, Kyunghwan Choi*
Withheld during double-blind review, 2026
1. Physics-Constrained Kalman Filtering for Wheel Normal Load Estimation Using Suspension and IMU Measurements
Angus Siegloff, **Myeongseok Ryu**, Kyunghwan Choi*
IEEE Transactions on Vehicle Technology, 2026

International Conference Papers

7. Constrained Optimization-Based Yaw-Rate Reference Map Generation for Active Rear Steering Control of Four-Wheel-Steering Vehicles
Myeongseok Ryu, Donghyun Hwang, Youngsik Yoon, Kyunghwan Choi*
International Conference on Control, Automation and Systems (ICCAS), (accepted, in press), 2026
6. Vector-Space Optimization Framework Based on Projected Contraction Condition for Control Design with Input Saturation
Myeongseok Ryu, Kyunghwan Choi*, Sesun You
International Federation of Automatic Control (IFAC) World Congress, (accepted, in press), 2026
5. All-Wheel Steering Vehicle Control Based on Contraction Theory with Neural Network
Myeongseok Ryu, Kyunghwan Choi*
IEEE International Conference on Advanced Motion Control (AMC), pp. 1-6, 2026
4. Physics-Informed Online Learning of Flux Linkage Model for Synchronous Machine
Seunghun Jang, **Myeongseok Ryu**, Kyunghwan Choi*
Annual Conference of the IEEE Industrial Electronics Society (IECON), pp. 1-7, 2025
3. Constrained Optimization-Based Neuro-Adaptive Control (CONAC) for Synchronous Machine Drives Under Voltage Constraints
Myeongseok Ryu, Niklas Monzen, Pascal Seitter, Kyunghwan Choi, Christoph M. Hackl*
Annual Conference of the IEEE Industrial Electronics Society (IECON), pp. 1-7, 2025
2. Imposing a Weight Norm Constraint for Neuro-Adaptive Control
Myeongseok Ryu, Jiyun Kim, Kyunghwan Choi*
European Control Conference (ECC), pp. 380-385, 2025
1. A Comparative Study of Reinforcement Learning and Analytical Methods for Optimal Control
Myeongseok Ryu, Junseo Ha, Minji Kim, Kyunghwan Choi*
International Workshop on Intelligent Systems (IWIS), pp. 1-5, 2023

Domestic Conference Papers

4. Constrained Optimization-Based Performance Improvement Analysis of Vehicle Lateral Control Using All-Wheel Steering
Myeongseok Ryu, Kyunghwan Choi*
제어로봇시스템학회 (ICROS), (accepted, in press), 2026
3. Approximation-based Steering Controller with Deep Neural Network
Myeongseok Ryu, Kyunghwan Choi*
제어로봇시스템학회 (ICROS), pp. 884-885, 2024
2. Integrated Motion Control of Four in-Wheel Motor Actuated Vehicles Considering Path Tracking, Ride Comfort, and Energy Efficiency
Myeongseok Ryu, Kyunghwan Choi*
한국자동차공학회 추계학술대회 (KSAE), pp. 490, 2023
1. Data-driven Modeling of Model Residuals for Linear Model Predictive Control of Nonlinear Systems
Myeongseok Ryu, Kyunghwan Choi*
제어로봇시스템학회 (ICROS), pp. 837-838, 2023

Preprint Papers

1. CNN-based End-to-End Adaptive Controller with Stability Guarantees
Myeongseok Ryu, Kyunghwan Choi*
Arxiv, 2024

GRANTS AND AWARDS

Best Presentation Paper Award

IEEE International Workshop on Intelligent Systems (IWIS)

July 2025

Student Support

European Control Association (EUCA)

June 2025

400 EUR

Research Support

Graduate International Research Experience Fellowship (GIST-IREF)

October 2024

16 million KRW (approx. 12,000 USD)

Best Paper Award*Institute of Control, Robotics and Systems (ICROS)**June 2023***INU MATLAB Cody Challenge***Top Prize**June 2021*